MV-PFAS-24-017 / Method conditions

North River Environmental Laboratory | EPA 533 modified | Revision A

QA REVIEWED

METHOD RECORD

Field	Recorded value
Instrument	SCIEX 6500+ QTRAP; ESI negative
Column	Acquity BEH C18, 2.1 x 50 mm, 1.7 um
Injection	5 uL
Mobile phase A	5 mM ammonium acetate in water
Mobile phase B	Methanol
Study dates	8-19 April 2024

ANALYTICAL WORKFLOW

Receipt → Preservation → SPE extraction → Dry-down → Reconstitution → LC-MS/MS → Review

BOTTLE AND QC MANIFEST

Sample ID	Matrix	Collected	Received	pH	Cl mg/L	mL	State
FB-240408-01	Field blank	08 Apr 09:10	08 Apr 15:42	6.8	<0.02	252	OK
DW-240408-02	Finished water	08 Apr 09:25	08 Apr 15:42	7.2	<0.02	251	OK
DW-240408-03	Finished water	08 Apr 09:40	08 Apr 15:42	7.1	<0.02	249	OK
DW-240409-04	Finished water	09 Apr 10:05	09 Apr 16:20	7.4	<0.02	250	OK
DW-240409-05	Finished water	09 Apr 10:20	09 Apr 16:20	7.3	<0.02	250	OK
LRB-240410-01	Reagent blank	10 Apr 07:30	10 Apr 07:30	6.9	<0.02	250	Prepared
LCS-240410-01	Control sample	10 Apr 07:35	10 Apr 07:35	7.0	<0.02	250	Prepared
MSD-240410-03	Matrix spike dup.	10 Apr 07:40	10 Apr 07:40	7.1	<0.02	250	Prepared

Calibration model and transitions

Quantifier/qualifier transitions, regression fields, and reporting limits

R = A_native / A_IS

C_sample = ((R - b) / m) x DF

LOD = t_(n-1,0.99) x SD_low

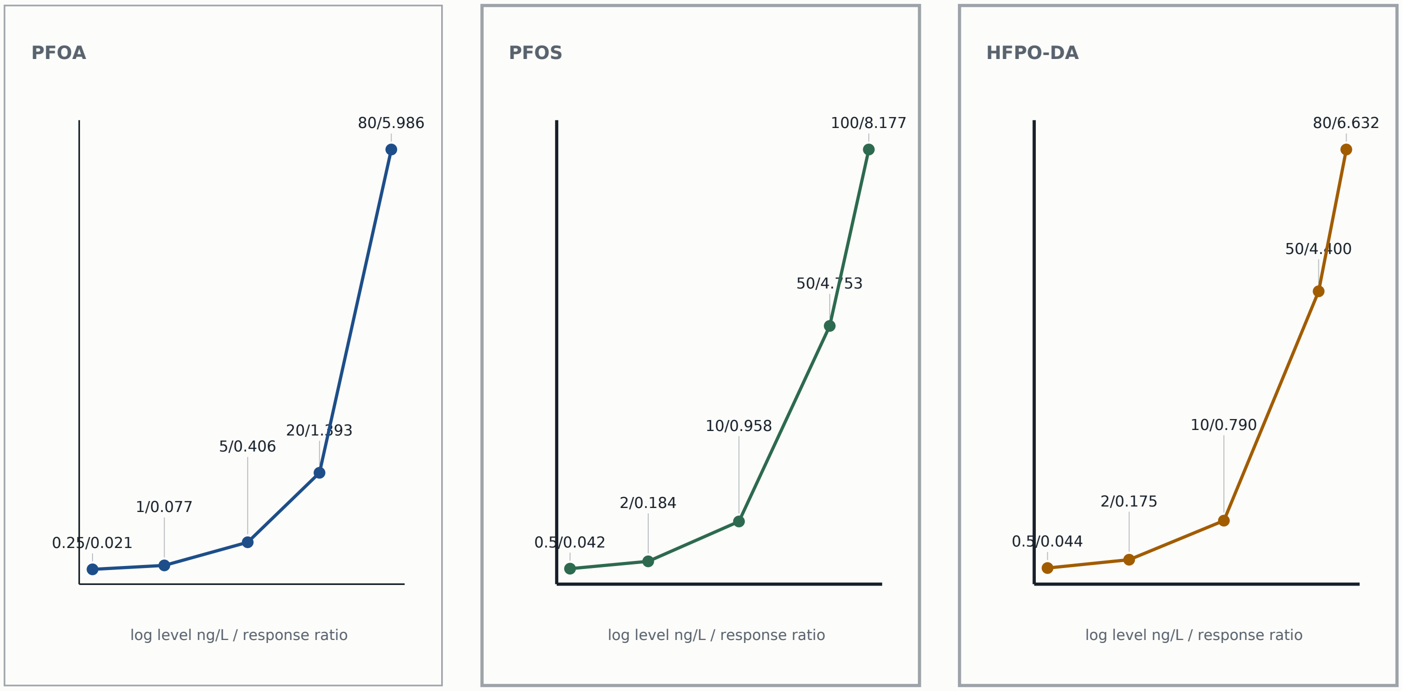
U95 = 2 x sqrt(u_cal^2 + u_rep^2 + u_vol^2 + u_rec^2)

TRANSITION AND CALIBRATION TABLE

Identity		Transitions		Retention	Regression	Fit	Limits ng/L
Analyte	**Internal standard**	**Quant.**	**Qual.**	**RT**	**m**	**R2**	**LOD / LOQ**
PFBA	13C4-PFBA	213>169	213>119	2.18	0.0921	0.9987	0.16 / 0.50
PFPeA	13C5-PFPeA	263>219	263>169	2.74	0.0875	0.9991	0.13 / 0.50
PFHxA	13C5-PFHxA	313>269	313>119	3.32	0.0814	0.9985	0.18 / 0.50
PFHpA	13C4-PFHpA	363>319	363>169	3.86	0.0778	0.9990	0.15 / 0.50
PFOA	13C8-PFOA	413>369	413>169	4.41	0.0749	0.9967	0.08 / 0.25
PFNA	13C9-PFNA	463>419	463>169	4.95	0.0712	0.9989	0.09 / 0.25
PFDA	13C6-PFDA	513>469	513>219	5.48	0.0686	0.9986	0.10 / 0.25
PFBS	18O2-PFBS	299>80	299>99	3.05	0.1033	0.9992	0.17 / 0.50
PFHxS	18O2-PFHxS	399>80	399>99	4.63	0.0968	0.9988	0.20 / 0.50
PFOS	13C8-PFOS	499>80	499>99	5.64	0.0915	0.9953	0.22 / 0.50
6:2 FTS	13C2-6:2FTS	427>407	427>81	5.19	0.0642	0.9979	0.31 / 1.00
HFPO-DA	13C3-HFPO-DA	329>285	329>169	3.71	0.0839	0.9979	0.14 / 0.50

Calibration curves and residual review

Five labeled levels per panel; weighted 1/x2 regression



RESIDUAL DISPOSITION

Analyte	Max abs residual	Failing points	Review decision
PFOA	7.8%	0	1/x2 retained
PFOS	10.6%	0	High calibrator inside +/-15%
HFPO-DA	6.1%	0	No curvature

Table S4 - Accuracy and precision

Part 1 of 2 | fortified reagent water | recovery % / RSD %

Analyte	Low rec / RSD	Mid rec / RSD	High rec / RSD	n	State
PFBA	96 / 8.4	101 / 4.6	99 / 3.8	7	Pass
PFPeA	93 / 9.1	98 / 5.2	101 / 4.1	7	Pass
PFHxA	91 / 10.3	97 / 5.8	100 / 4.3	7	Pass
PFHpA	95 / 7.9	99 / 4.9	102 / 4.0	7	Pass
PFOA	98 / 6.8	103 / 3.7	101 / 3.2	7	Pass
PFNA	97 / 7.2	102 / 4.1	100 / 3.6	7	Pass

ACCEPTANCE CRITERIA

Recovery 70-130%; RSD <= 20%. Low level equals 2x LOQ for PFOA, PFNA, and PFDA and equals LOQ for the remaining analytes.

Table S4 continued — PFDA through HFPO-DA appear on page 5.

Table S4 - Accuracy and precision (continued)

Part 2 of 2 | fortified reagent water | recovery % / RSD %

Analyte	Low rec / RSD	Mid rec / RSD	High rec / RSD	n	State
PFDA	92 / 11.5	96 / 6.3	99 / 4.9	7	Pass
PFBS	104 / 9.7	99 / 4.8	98 / 4.0	7	Pass
PFHxS	101 / 8.9	100 / 4.4	97 / 3.9	7	Pass
PFOS	89 / 12.6	95 / 6.7	96 / 5.1	7	Pass
6:2 FTS	86 / 14.8	92 / 7.4	94 / 6.3	7	Pass
HFPO-DA	99 / 7.5	104 / 4.2	103 / 3.5	7	Pass

DW-240408-03 MATRIX SPIKE DUPLICATE

Analyte	Native	Spike	MS rec	MSD rec	RPD	Qualifier
PFOA	3.42	10.00	96	98	2.1	None
PFOS	0.74	10.00	91	88	3.4	J: native <2x LOQ
PFHxS	1.16	10.00	102	99	3.0	None
HFPO-DA	<0.50	10.00	105	106	1.0	U native
6:2 FTS	<1.00	10.00	84	87	3.5	Low bias monitored

Instrument sequence and reinjection review

Signed bench copy scanned after technical review

LC-MS/MS BATCH SEQUENCE / MV-PFAS-24-017

Seq	Type	Vial	Sample ID	Time	IS	Carry	State
01	Blank	A01	SBLK-01	08:10	Y	Clear	Accepted
02	Cal	A02	CAL-0.5	08:23	Y	Clear	Accepted
03	Cal	A03	CAL-1	08:36	Y	Clear	Accepted
04	Cal	A04	CAL-2	08:49	Y	Clear	Accepted
05	Cal	A05	CAL-5	09:02	Y	Clear	Accepted
06	Cal	A06	CAL-10	09:15	Y	Clear	Accepted
07	Cal	A07	CAL-20	09:28	Y	Clear	Accepted
08	Cal	A08	CAL-50	09:41	Y	Clear	Accepted
09	Cal	A09	CAL-80	09:54	Y	Clear	Accepted
10	Blank	B01	LRB-240410-01	10:07	Y	Clear	Accepted
11	LCS	B02	LCS-240410-01	10:20	Y	Clear	Accepted
12	Sample	B03	FB-240408-01	10:33	Y	Clear	Accepted
13	Sample	B04	DW-240408-02	10:46	Y	Clear	Accepted
14	Sample	B05	DW-240408-03	10:59	Y	Clear	Accepted
15	MS	B06	DW-240408-03-MS	11:12	Y	Clear	Accepted
16	MSD	B07	DW-240408-03-MSD	11:25	Y	Clear	Accepted
17	CCV	B08	CCV-10	11:38	Y	Clear	Accepted
18	Sample	B09	DW-240409-04	11:51	N	Clear	Reinject
19	Sample	C01	DW-240409-04-RI	12:18	Y	Clear	Accepted
20	Sample	C02	DW-240409-05	12:31	Y	Clear	Accepted

Analyst M. Iyer 19 Apr 14:05 Technical L. Chen 22 Apr 09:40

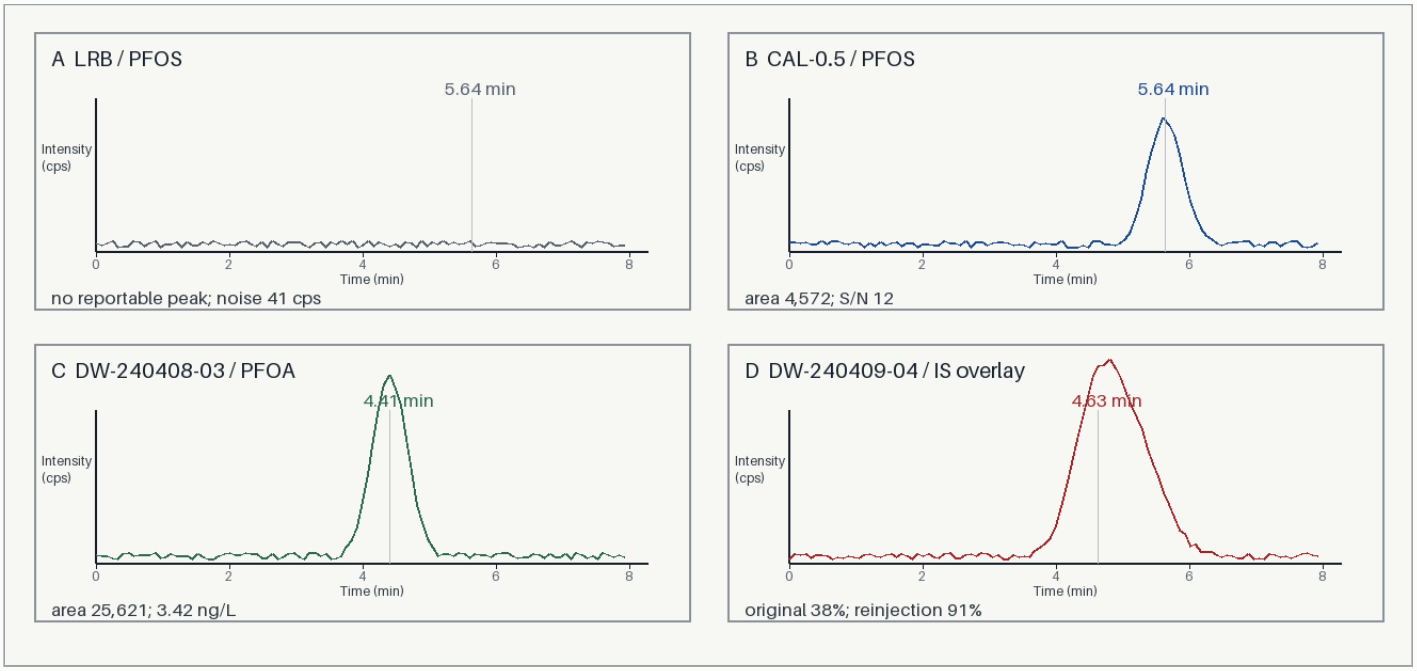
QA R. Patel 22 Apr 16:10 Seq 19 controls DW-240409-04 results

✓

Sequence 18 failed internal-standard recovery. Sequence 19 is the accepted reinjection and supplies the final DW-240409-04 result.

Chromatograms, final results, and uncertainty

Plate S7; references distinguish direct traces from contextual QC evidence



Result ID	Sample	Analyte	Result	Qual.	U95	Reference
R-01	FB-240408-01	PFOA	<0.25	U	-	p2 limit / context
R-02	DW-240408-02	PFOA	2.18	-	0.42	S7C / other sample
R-03	DW-240408-02	PFOS	0.62	J	0.18	S7B / cal context
R-04	DW-240408-03	PFOA	3.42	-	0.55	S7C / direct
R-05	DW-240408-03	PFOS	0.74	J	0.19	S7A / blank context
R-06	DW-240409-04	PFOA	1.06	J	0.25	S7D IS / Seq 19
R-07	DW-240409-04	PFOS	<0.50	U	-	S7D IS / Seq 19
R-08	DW-240409-05	PFOA	4.87	-	0.72	No S7 panel

EXPANDED UNCERTAINTY

Analyte	u cal	u rep	u vol	u rec	U95
PFOA	5.2%	4.1%	1.5%	3.6%	15.4%
PFOS	7.4%	6.9%	1.5%	5.1%	22.8%
HFPO-DA	4.8%	3.7%	1.5%	3.2%	13.8%